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TRƯỜNG ĐẠI HỌC KINH TẾ
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From Retrospective Benchmarking to Anticipatory Policy: A Data-Driven Framework for Governance Indexing and Forecasting Using the Viet Nam Provincial Governance and Public Administration Performance Index (PAPI)

PROJECT ID: XHNV.17.01
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1. INTRODUCTION

PROBLEM STATEMENT

- Current subnational governance indices (PAPI, PCI) are predominantly retrospective and rely on equal-weighting defaults.
- This limits capacity to provide forward-looking signals and early warnings for proactive policymaking.
- A data-driven, anticipatory framework is urgently needed across Vietnam's 63 provinces.

RESEARCH OBJECTIVES

1. Develop a hybrid governance framework combining Two-Level Hierarchical CRITIC Weighting with multi-window temporal stability validation (RMSD, Spearman's ρ, Kendall's W, Monte Carlo sensitivity).
2. Establish a multi-methodological MCDM Consensus Ranking framework (TOPSIS, VIKOR, PROMETHEE II, COPRAS, EDAS) to transcend single-method idiosyncrasy, validate inter-method concordance, and identify an optimal ranking function.
3. Generate probabilistic projections for 2025 to operationalize an institutional Early Warning System for subnational governance.

2. METHODS & DATA

DATA DESCRIPTION

- Balanced panel dataset from PAPI (2011–2024) across all 63 provinces of Viet Nam.
- 2011–2017: 6 criteria (C01–C06), comprising 21 sub-criteria.
- 2018–2024: Expanded to 8 criteria, adding C07: Environmental Governance and C08: E-Governance, totalling 29 sub-criteria (SC11–SC83).

THREE-TIER METHODOLOGICAL FRAMEWORK

PAPI Longitudinal Panel 2011–2024

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TIER 1 — Two-Level Hierarchical CRITIC Weighting
Level 1: Sub-criterion weights within criteria · Level 2: Cross-criteria weights capturing contrast intensity

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TIER 2 — Multi-Method MCDM Consensus Ranking
TOPSIS · VIKOR · PROMETHEE II · COPRAS · EDAS

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TIER 3 — AutoGluon Stacked Ensemble Forecasting (2025)
10 diverse base learners: Statistical · Tabular ML · Deep Learning · Chronos foundation model

3. WEIGHTING & RANKING RESULTS

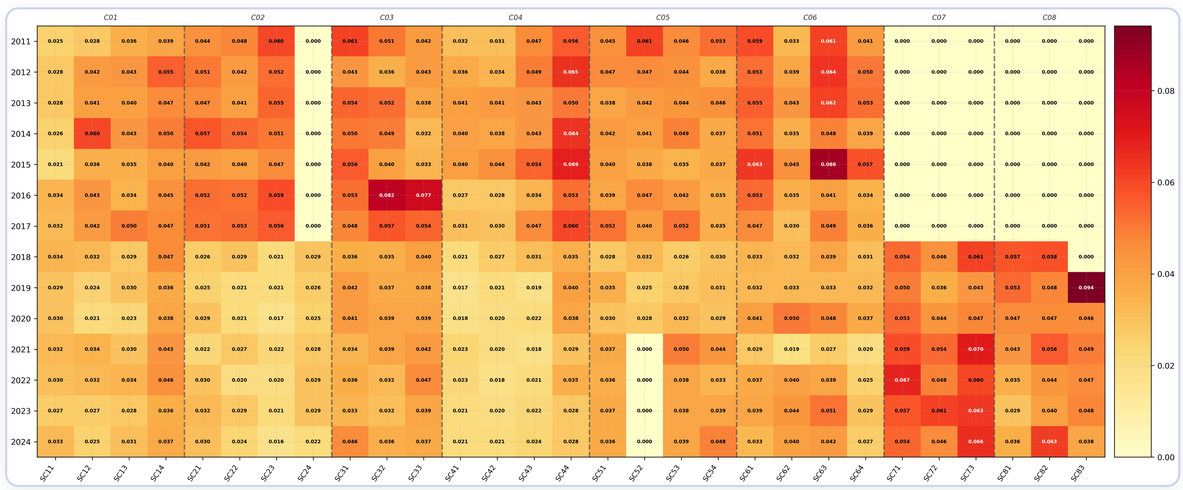


Fig. 1 — Sub-criterion weight heatmap (2011–2024)

TABLE 1 — MEAN CRITIC CRITERION WEIGHTS

CRITERION	MEAN WT.	CV	INFORMATIONAL ROLE
C01: Local Participation	14.21%	0.149	Lowest weight variance (most consistent)
C02: Transparency	12.46%	0.230	Lowest weight among long-standing criteria
C03: Vertical Accountability	13.22%	0.211	Moderate information value
C04: Control of Corruption	13.64%	0.317	Highest temporal weight volatility
C05: Public Admin. Procedures	14.59%	0.213	Highly stable administrative priority
C06: Public Service Delivery	16.76%	0.238	Primary differentiator across provinces
C07: Env. Governance *(since 2018)*	16.28%	0.142	Graceful integration of index expansion
C08: E-Governance *(since 2018)*	13.99%	0.157	Graceful integration of index expansion

KEY FINDINGS & VALIDATION

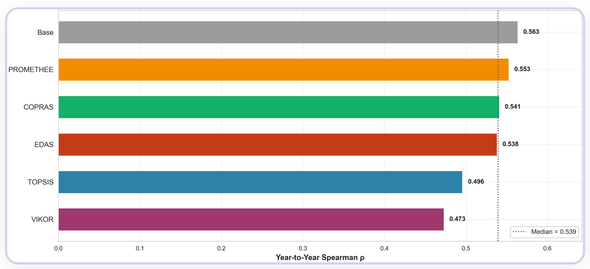


Fig. 2 — Rank stability over time

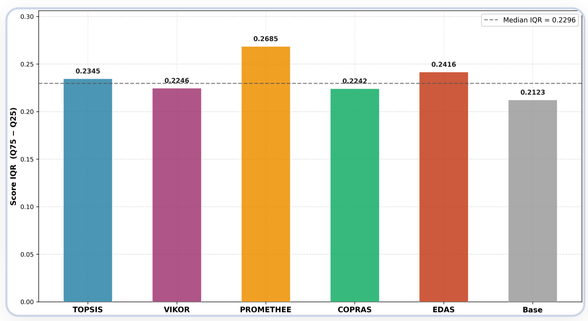


Fig. 3 — Ranking discriminatory power (Score IQR)

4. FORECASTING RESULTS (2025)

0.6411 — Ensemble MASE — *Well below 1.0 naive baseline*
+14.12% — vs. Best Single (TFT) — *TFT baseline MASE = 0.7316*
+86.0% — vs. Naive Baseline — *Naive MASE = 1.1924*

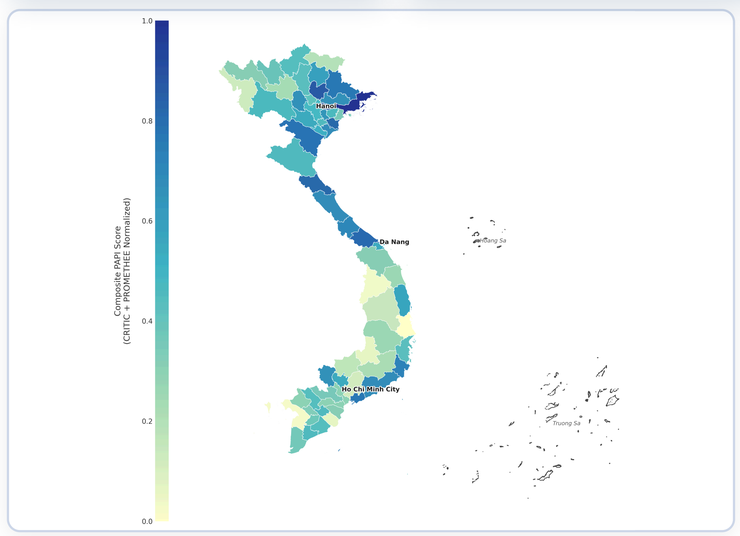


Fig. 4 — 2025 Provincial Governance Forecast (PROMETHEE II scores)

LEADING CLUSTER

Quang Ninh	1.000
Thai Nguyen	0.862
Bac Ninh	0.855
Ha Tinh	0.819

LAGGING CLUSTER

Phu Yen	0.000
Kien Giang	0.020
Kon Tum	0.037
Dak Nong	0.060

METROPOLITAN

Hanoi	0.477
Ho Chi Minh	0.302

Moderate scores reflect urbanisation-driven service scaling challenges.

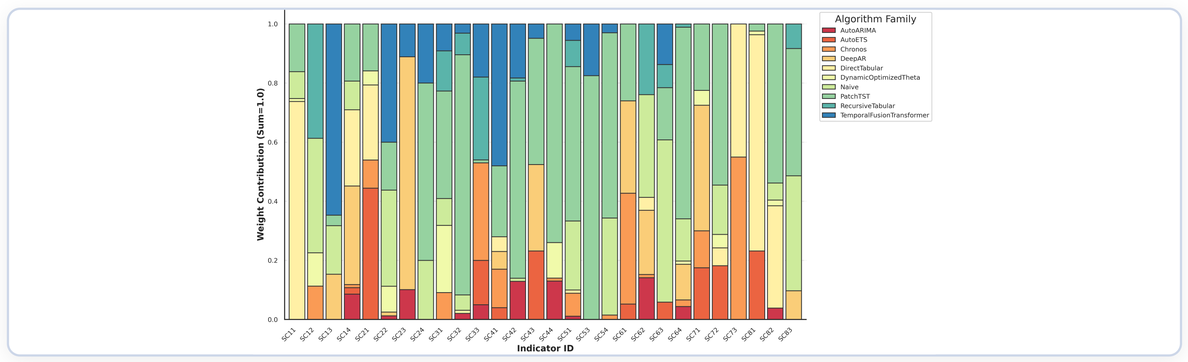


Fig. 5 — AutoGluon stacked ensemble model family weight distribution

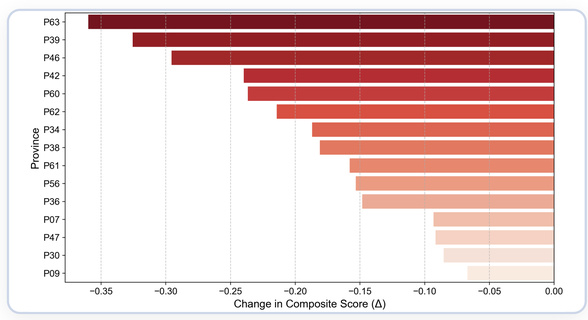


Fig. 6 — Composite PAPI score decline forecast

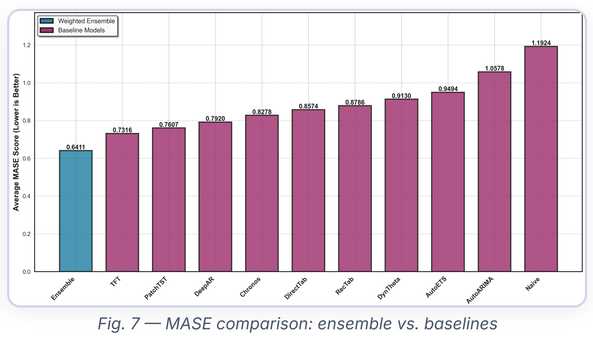


Fig. 7 — MASE comparison: ensemble vs. baselines

5. DISCUSSION & POLICY IMPLICATIONS

PROJECTED GOVERNANCE CONTRACTIONS

- 29 of 63 provinces are projected to experience governance contractions in 2025.
- Systemic bottlenecks: Internet Access (SC82) and Civic Knowledge (SC11) signal issues in digital administration and grassroots civic engagement.
- Top declines: Ca Mau (~50%), Dak Lak (~48%), Quang Ngai (41.12%), Binh Thuan (32.55%).

5-PILLAR POLICY RESPONSE MATRIX

POLICY PILLAR	TARGET SUB-CRITERIA	REGULATORY & OPERATIONAL RESPONSE
1 Digital Access & E-Governance	SC82 Internet Access SC81 E-Gov Portals	– Ringfence USO capital for last-mile broadband in lagging provinces. – Align KPIs with Nat. Digital Transformation Program (Decision 749/QD-TTg). – Deploy commune-level digital navigators (Lao Cai model).
2 Civic Engagement & Grassroots Democracy	SC11 Civic Knowledge SC14 Voluntary Contrib.	– Mandate commune deliberative forums (Law 10/2022/QH15, Decree 59/2023). – Establish Village Development Funds with transparent community auditing.
3 Accountability & Anti-Corruption	SC23 Budget Trans. SC42 Corruption Control	– Standardize commune budget reports via MoF templates. – Deploy targeted SAV audits in high-risk provinces. – Mandate e-procurement at district level.
4 Environmental & Resource Governance	SC73 Water Quality SC71 Env. Protection	– Inter-provincial Mekong water quality compact to resolve jurisdictional gaps. – Recapitalize forest payments (Decree 156/2018 & 91/2024) in Central Highlands.
5 Tiered Governance Compact (TGC)	Composite Scores	– Top-Quartile Provinces: Enhanced fiscal autonomy and reduced compliance burden. – Bottom-Quartile / Declining: Trigger central support mechanisms and capacity programs.

6. CONCLUSION

ACADEMIC CONTRIBUTION
Bridges static MCDM ranking with dynamic temporal ML forecasting, defining a replicable methodology for Anticipatory Governance in developing economies.

POLICY VALUE
Transitions subnational governance from retrospective benchmarking to proactive one-year-ahead early-warning interventions at scale.

REPRODUCIBILITY
Code repositories and processed datasets are published as open-source assets for full scientific reproducibility and peer replication.

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